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An Analysis of Security Risks and Benefits of Cloud Computing for Organizations

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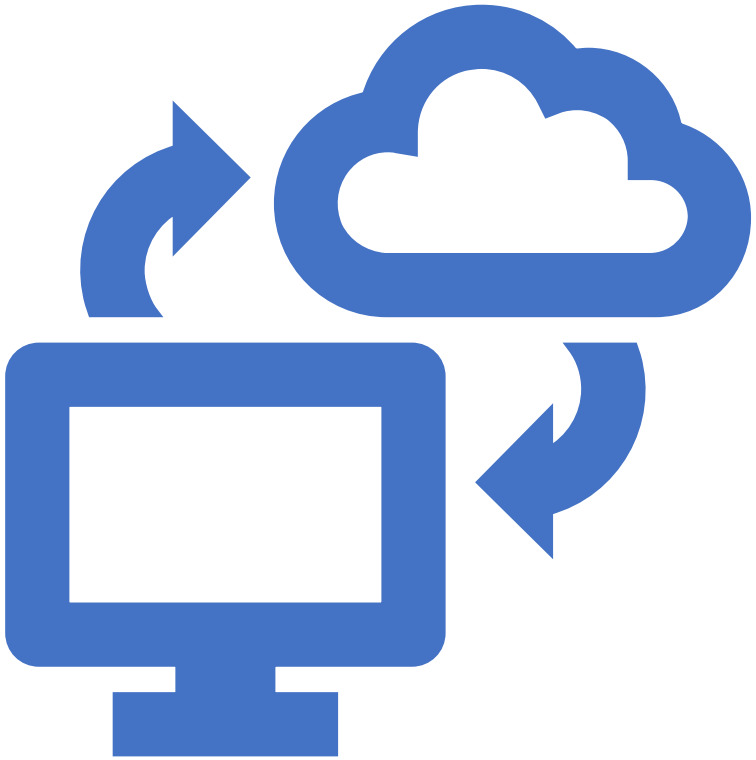


An Analysis of Security Risks and Benefits of Cloud Computing for Organizations.

Introduction to cloud computing

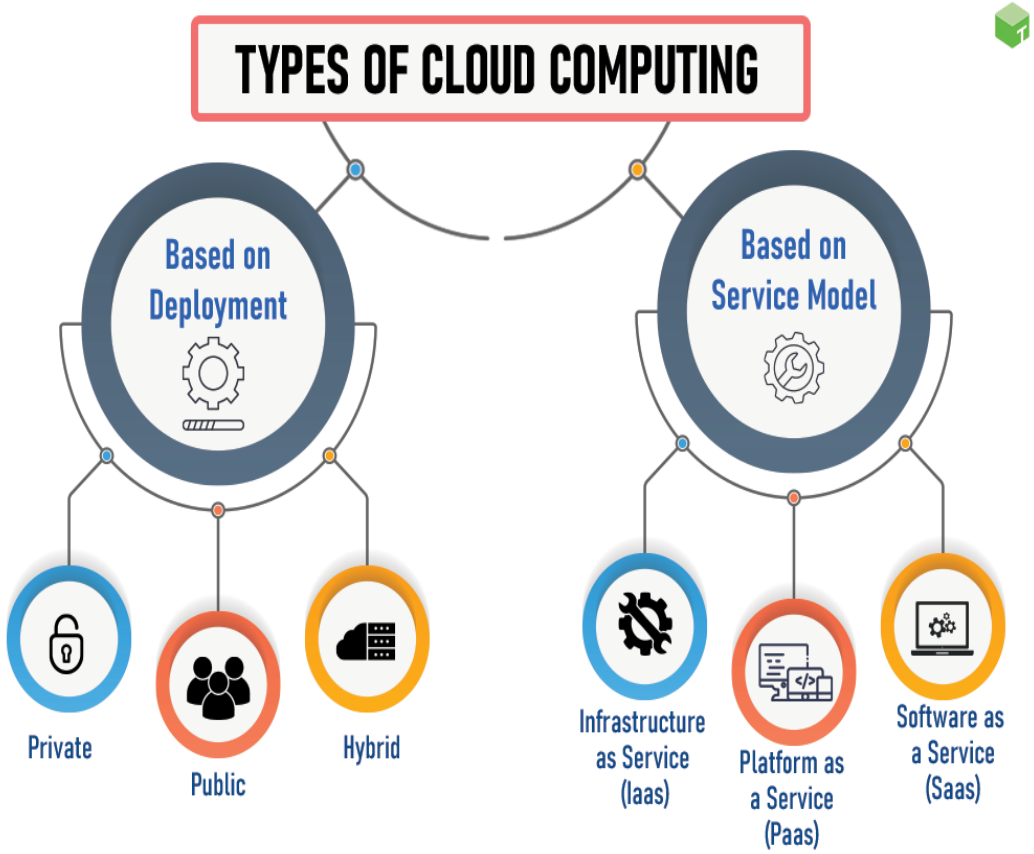
- Cloud computing is defined as the use of hosted services, such as data storage, servers, databases, networking, and software over the internet.
- The data is stored on physical servers, which are maintained by a cloud service provider.
- Computer system resources, especially data storage and computing power, are available on-demand, without direct management by the user in cloud computing.

Key Characteristics of Cloud Computing

There are 5 key characteristics of cloud computing environment, as defined by the National Institute of Standards and Technology (NIST):

- **Internet Access**
- **Measured Service**
- **On-Demand Self-Service**
- **Shared Resource Pooling**
- **Rapid Elasticity**

Types of Cloud Computing



- Cloud computing can either be classified based on the deployment model or the type of service.
- Based on the specific deployment model, we can classify cloud as **public, private, and hybrid cloud**.
- At the same time, it can be classified as infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and software-as-a-service (SaaS) based on the service the cloud model offers.

Deployment models

1. **Public Cloud**

- Public cloud refers to computing services offered by third-party providers over the internet.

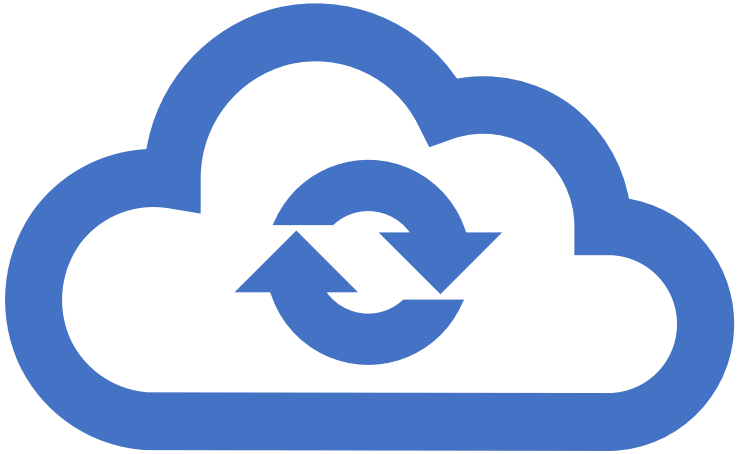
- Public clouds can help businesses save on purchasing, managing, and maintaining on-premises infrastructure since the cloud service provider is responsible for managing the system.

The largest cloud service providers with data centers that enable massive scaling are called hyperscalers.

- The big four hyperscalers, collectively referred to as MAAGs, are:

1. Microsoft (Azure),
2. Amazon (Amazon Web Services or AWS),
3. Alibaba (Alicloud),
4. Google (Google Cloud).

- Other cloud providers include IBM and Oracle.



2. Private cloud:

- In a private cloud, the computing services are offered over a private IT network for the dedicated use of a single organization.
- Also termed internal, enterprise, or corporate cloud, a private cloud is usually managed via internal resources and is not accessible to anyone outside the organization.
- Private cloud computing provides all the benefits of a public cloud, such as self-service, scalability, and elasticity, along with additional control, security, and customization.

Benefit and Drawback of Private Cloud

- provides a higher level of security through company firewalls and internal hosting

- the organization becomes responsible for all the management and maintenance of the data centers

3. Hybrid cloud

- Hybrid cloud uses a combination of public and private cloud features.

- The “best of both worlds” cloud model allows a shift of workloads between private and public clouds as the computing and cost requirements change.

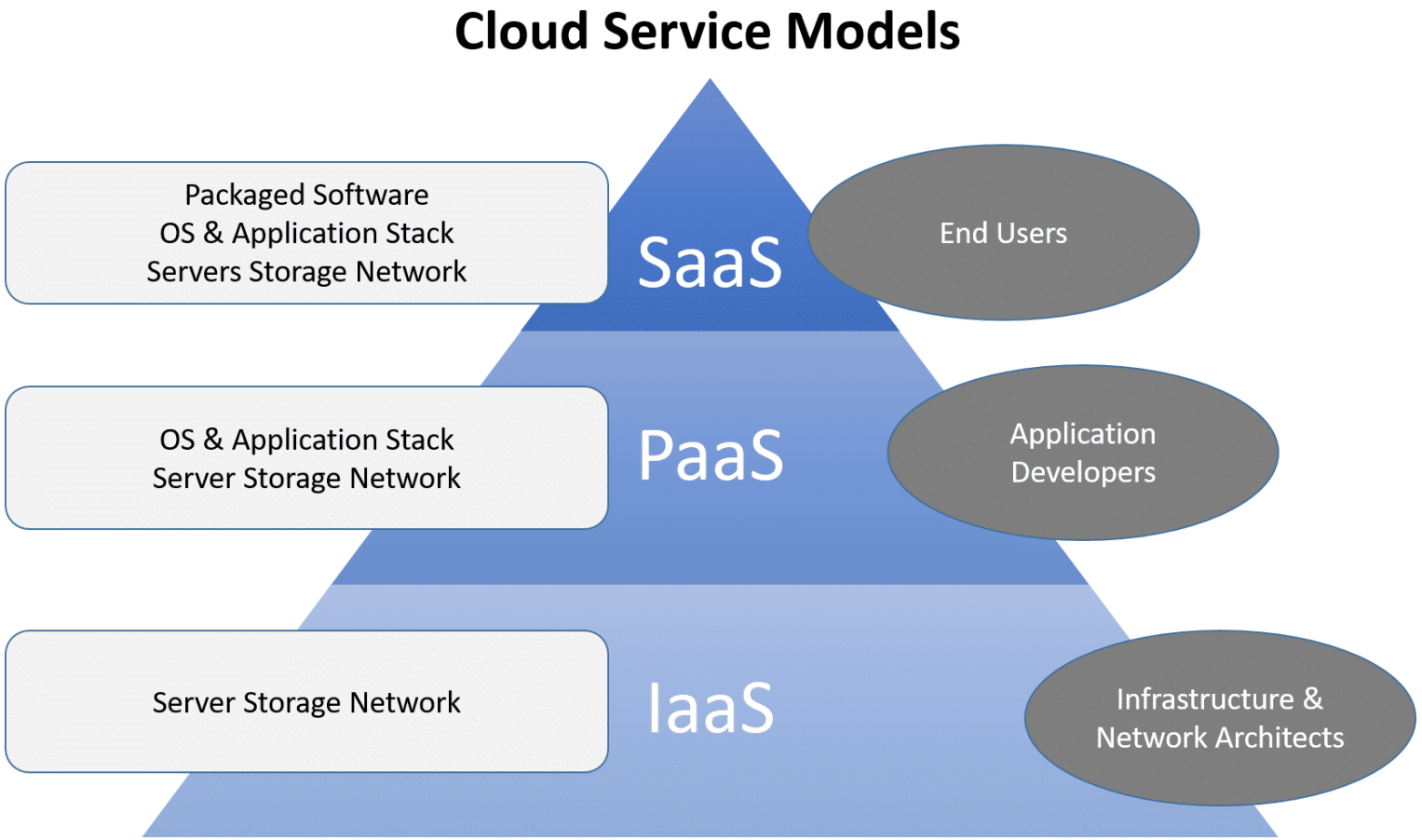
- Many organizations actually use a combination of several cloud environments.

- Hybrid cloud eliminates reliance on any single cloud provider and allows for additional levels of flexibility in terms of capabilities, security compliance, etc.

Multi-cloud

A multi-cloud approach is a particular case of hybrid cloud in which an organization uses services from multiple public cloud providers.

Service categories



The 6 advantages of Cloud Computing

The 6 advantages of Cloud Computing

1. Trade capital expense for variable expense.

Purchase servers.

Tax deductible over depreciation lifetime.

Pay as you go.

Tax deductible in the same year.

The 6 advantages of Cloud Computing

2. Benefit from massive economies of sale

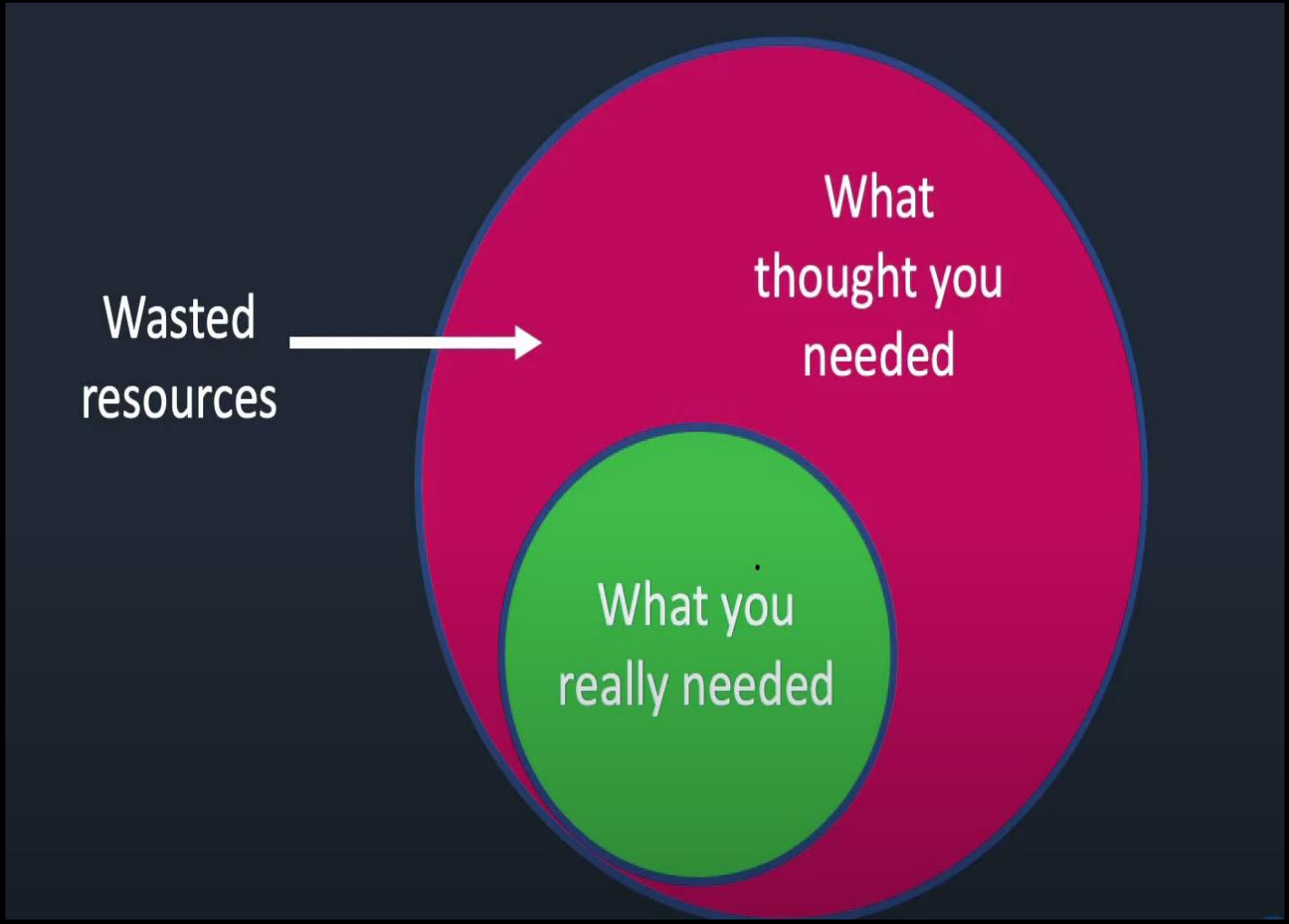
The 6 advantages of Cloud Computing

2. Benefit from massive economies of scale

- Aggregate usage across hundreds of thousands of customers = lower variable costs for customers

The 6 advantages of Cloud Computing

3. Stop guessing capacity.



The 6 advantages of Cloud Computing

4. Increase speed and agility

Speed = deploy resources easily and quickly

Agility = react to change ; speed to market

The 6 advantages of Cloud Computing

5. Stop spending money running and maintaining data centers

The 6 advantages of Cloud Computing

6. Go global in minutes

Security risks of cloud computing

1. **Misconfiguration issues**

- A small misconfigured component can have a big impact on cloud security and lead to a ripple effect on the security issues in other assets.

-Few components that lead to the misconfiguration security issue in cloud computing.

- Unrestricted outbound access
- Privileged access posing security risks in cloud computing environments
- Unencrypted tokens or keys

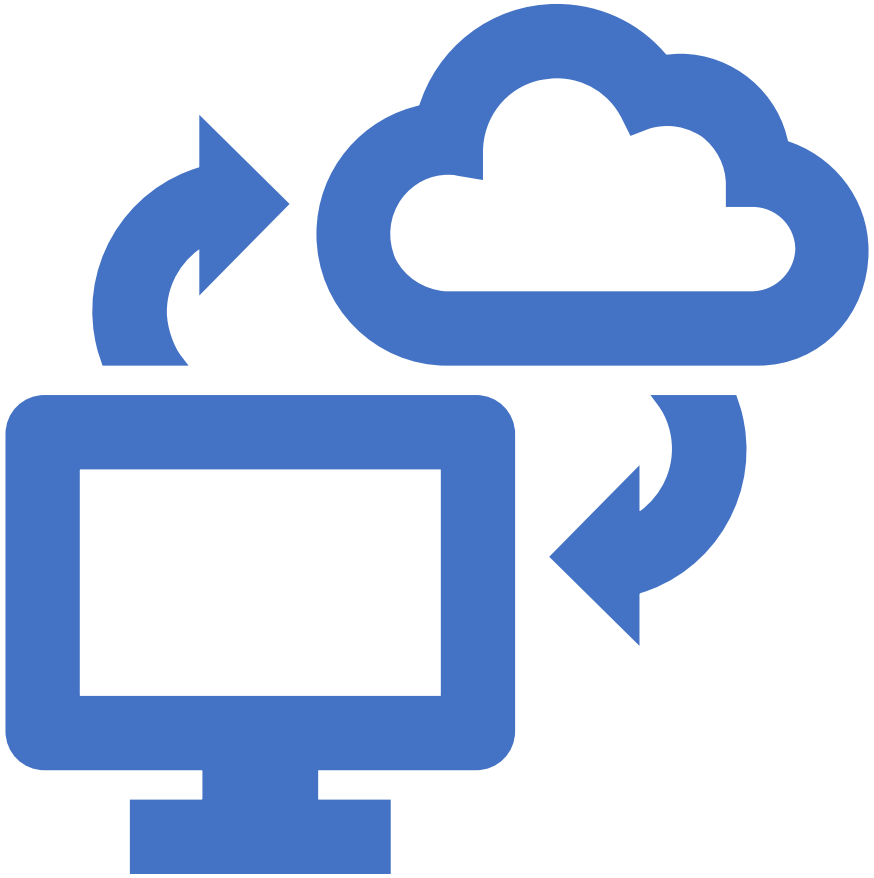
2. **Insecure API**

In cloud computing security, the APIs are the backbone of processing and are designed to facilitate data access and integration.

The primary reason for the insecure API layer is improper cloud security design policies, lack of authentication and access control mechanism with wrong dependency chain of APIs.

3. Data sharing

- People can share data and files externally via email or a publicly accessible link in a cloud ecosystem.
- This feature for reaching out to the file with the URL shared on public visibility can have multiple security risks on cloud computing assets
- Any attacker who gets the URL by any cyber attack or breach can invade the shared data privacy, confidentiality, and integrity.



4. **Data Loss or theft**

- Cloud-based file sharing and storing services are accessible via the public internet and contain important confidential information not meant for public release.

- Suppose any attack befalling the cloud provider or the customer through any security attack can jeopardize the whole data confidentiality.

5. **Data privacy compliances**

- Data protection rules and regulations such as PCI DSS, HIPAA, GDPR, etc., take the organisation liable for processing and protecting the customer’s personal data stored in the cloud.

- Every information stored and accessible online takes a minor glitch to cause a data breach and violate the compliance requirements.

6. **Lack of access control policies**

- Due to a lack of user access control policies, there are high chances of data theft or data leak.

- With appropriate access control policies and monitoring, businesses can regulate permission and reduce the cause of data breaches or theft costs that may impact the business in terms of time, money, or reputation.

7. Denial of Service (DoS/DDoS)

8. Accidental credential exposure

9. Insider Threats

10. Malware

Mitigation of security risks in cloud computing

1. **Regular third-party checks**

It is important to identify gaps in the security controls regularly.

Cloud security assessments such as cloud penetration testing, AWS security testing, Azure testing or SaaS security assessments should be undertaken regularly.

2. **Secure hardening practices**

- Ensure technical security baselines are in place and secure hardening ensures security controls before any assets are released in the production environment.

3. Multi-factor authentication

- Two-factor authentication (2FA) must be added as an extra authentication factor to access cloud accounts.

- This will help reduce the likelihood of account compromises by a great deal, whether it is an insider attack or a credential leakage concern.

4. **Secure backups**

- It is imperative that you routinely perform secure backups (at data centres or separate cloud storage locations) to ensure your files are not lost.

- This may be provided as a feature by your cloud service provider.

5. **User training**

- Training your staff regularly on the digital risks, cloud security concerns, in this case, helps to ensure staff understand the need to protect assets from accidental events such as phishing attacks and social engineering attacks.

6. **Encryption**

Conclusion

- You need to be proactive in identifying all potential threats before they happen so that you can take preventative measures and mitigate risk.